```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

enrollment=pd.read_csv('W.Champaran Aadhar Enrollment Dataset.csv')

enrollment

           date  state            district  pincode  age_0_5  age_5_17  \
0    01-06-2025  Bihar  Pashchim Champaran   845450       88       262   

1    01-06-2025  Bihar  Pashchim Champaran   845106      197       333   

2    01-07-2025  Bihar  Pashchim Champaran   845451      202       354   

3    01-07-2025  Bihar  Pashchim Champaran   845306      189       436   

4    01-07-2025  Bihar  Pashchim Champaran   845438      608      1229   

..          ...    ...                 ...      ...      ...       ...   

101  23-12-2025  Bihar  Pashchim Champaran   845105        2         0   

102  10-06-2025  Bihar  Pashchim Champaran   845455        1         1   

103  12-06-2025  Bihar  Pashchim Champaran   845438        2         0   

104  17-06-2025  Bihar  Pashchim Champaran   845106        0         1   

105  19-06-2025  Bihar  Pashchim Champaran   845455        2         0   


     age_18_greater
0                10
1                13
2                30
3                33
4                34
..              ...
101               0
102               0
103               0
104               0
105               0

[106 rows x 7 columns]

enrollment.head()
```

```
         date  state           district  pincode  age_0_5  
age_5_17  \
0  01-06-2025  Bihar  Pashchim Champaran   845450       88       262   

1  01-06-2025  Bihar  Pashchim Champaran   845106      197       333   

2  01-07-2025  Bihar  Pashchim Champaran   845451      202       354   

3  01-07-2025  Bihar  Pashchim Champaran   845306      189       436   

4  01-07-2025  Bihar  Pashchim Champaran   845438      608      1229   


   age_18_greater  
0              10  
1              13  
2              30  
3              33  
4              34  
```

```
enrollment.tail()
```

```
           date  state           district  pincode  age_0_5  age_5_17
\
101  23-12-2025  Bihar  Pashchim Champaran   845105        2         0   

102  10-06-2025  Bihar  Pashchim Champaran   845455        1         1   

103  12-06-2025  Bihar  Pashchim Champaran   845438        2         0   

104  17-06-2025  Bihar  Pashchim Champaran   845106        0         1   

105  19-06-2025  Bihar  Pashchim Champaran   845455        2         0   


     age_18_greater  
101               0  
102               0  
103               0  
104               0  
105               0  
```

```
enrollment.shape
```

```
(106, 7)
```

```
enrollment.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106 entries, 0 to 105
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype 
```

```
---  ------          --------------  -----
 0   date            106 non-null    object
 1   state           106 non-null    object
 2   district        106 non-null    object
 3   pincode         106 non-null    int64
 4   age_0_5         106 non-null    int64
 5   age_5_17        106 non-null    int64
 6   age_18_greater  106 non-null    int64
dtypes: int64(4), object(3)
memory usage: 5.9+ KB
```

```
enrollment.isnull().head()
```

```
    date  state  district  pincode  age_0_5  age_5_17  age_18_greater
0  False  False     False    False    False     False           False
1  False  False     False    False    False     False           False
2  False  False     False    False    False     False           False
3  False  False     False    False    False     False           False
4  False  False     False    False    False     False           False
```

```
enrollment.duplicated()
```

```
0      False
1      False
2      False
3      False
4      False
       ...
101    False
102    False
103    False
104    False
105    False
Length: 106, dtype: bool
```

```
enrollment.head()
```

```
         date  state            district  pincode  age_0_5
age_5_17  \
0  01-06-2025  Bihar  Pashchim Champaran   845450       88        262

1  01-06-2025  Bihar  Pashchim Champaran   845106      197        333

2  01-07-2025  Bihar  Pashchim Champaran   845451      202        354

3  01-07-2025  Bihar  Pashchim Champaran   845306      189        436

4  01-07-2025  Bihar  Pashchim Champaran   845438      608       1229


   age_18_greater
```

```
0               10
1               13
2               30
3               33
4               34
```

```
enrollment['age_18_greater'].value_counts().head(5)
```

```
age_18_greater
0     84
13     2
17     2
26     2
33     1
Name: count, dtype: int64
```

```
enrollment['date'] = pd.to_datetime(enrollment['date'], format='%d-%m-
%Y')
```

```
enrollment.head()
```

```
        date            district  pincode  age_0_5  age_5_17
age_18_greater
0 2025-06-01  Pashchim Champaran   845450       88       262
10
1 2025-06-01  Pashchim Champaran   845106      197       333
13
2 2025-07-01  Pashchim Champaran   845451      202       354
30
3 2025-07-01  Pashchim Champaran   845306      189       436
33
4 2025-07-01  Pashchim Champaran   845438      608      1229
34
```

```
enrollment.describe()
```

```
                                date        pincode     age_0_5
age_5_17  \
count                            106     106.000000  106.000000
106.000000
mean   2025-09-05 05:53:12.452830208  845328.103774   47.830189
101.481132
min              2025-03-27 00:00:00  845101.000000    0.000000
0.000000
25%              2025-06-17 06:00:00  845105.000000    0.000000
0.000000
50%              2025-09-12 12:00:00  845438.000000    1.000000
1.000000
75%              2025-11-07 18:00:00  845452.000000    2.000000
2.000000
max              2026-01-03 00:00:00  845459.000000  608.000000
```

```
1229.000000
std                                 NaN     159.537840  106.712740
224.530165

       age_18_greater
count      106.000000
mean         5.301887
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max         44.000000
std         11.472675
```

```
enrollment[['age_0_5', 'age_5_17', 'age_18_greater']].describe()
```

	age_0_5	age_5_17	age_18_greater
count	106.000000	106.000000	106.000000
mean	47.830189	101.481132	5.301887
std	106.712740	224.530165	11.472675
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	1.000000	1.000000	0.000000
75%	2.000000	2.000000	0.000000
max	608.000000	1229.000000	44.000000

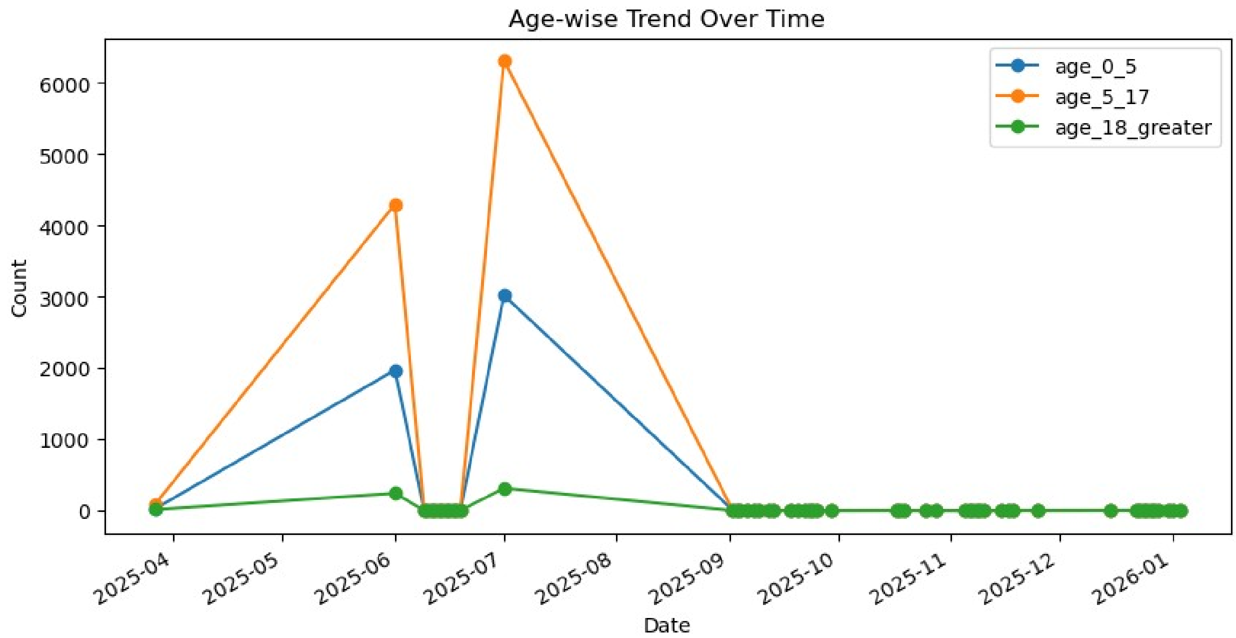
```
enrollment.groupby('date')[['age_0_5', 'age_5_17',
'age_18_greater']].sum()
```

date	age_0_5	age_5_17	age_18_greater
2025-03-27	25	88	13
2025-06-01	1969	4299	239
2025-06-09	0	2	0
2025-06-10	2	2	0
2025-06-11	0	1	0
2025-06-12	3	0	0
2025-06-13	1	0	0
2025-06-14	0	2	0
2025-06-15	0	1	0
2025-06-16	1	2	0
2025-06-17	0	2	0
2025-06-18	1	1	0
2025-06-19	2	1	0
2025-07-01	3018	6314	310
2025-09-02	3	0	0
2025-09-03	3	2	0
2025-09-04	2	1	0
2025-09-06	1	0	0
2025-09-08	1	0	0

```
2025-09-09        2         2              0
2025-09-12        2         0              0
2025-09-13        1         2              0
2025-09-18        2         1              0
2025-09-20        1         1              0
2025-09-22        1         0              0
2025-09-23        0         1              0
2025-09-24        1         1              0
2025-09-25        0         1              0
2025-09-29        1         0              0
2025-10-17        2         2              0
2025-10-18        2         0              0
2025-10-19        0         2              0
2025-10-25        0         2              0
2025-10-28        0         1              0
2025-11-05        1         0              0
2025-11-06        1         0              0
2025-11-07        1         0              0
2025-11-08        0         3              0
2025-11-09        1         1              0
2025-11-10        1         0              0
2025-11-15        2         0              0
2025-11-17        0         1              0
2025-11-18        3         1              0
2025-11-25        2         0              0
2025-12-15        2         3              0
2025-12-22        4         0              0
2025-12-23        3         0              0
2025-12-24        0         1              0
2025-12-25        0         1              0
2025-12-26        0         1              0
2025-12-27        0         1              0
2025-12-28        1         0              0
2025-12-31        1         5              0
2026-01-01        0         4              0
2026-01-03        0         1              0
```

Age-wise Trend

```
enrollment.groupby('date')[['age_0_5', 'age_5_17',
'age_18_greater']].sum().plot(
    figsize=(10,5), marker='o'
)
plt.title("Age-wise Trend Over Time")
plt.ylabel("Count")
plt.xlabel("Date")
plt.show()
```



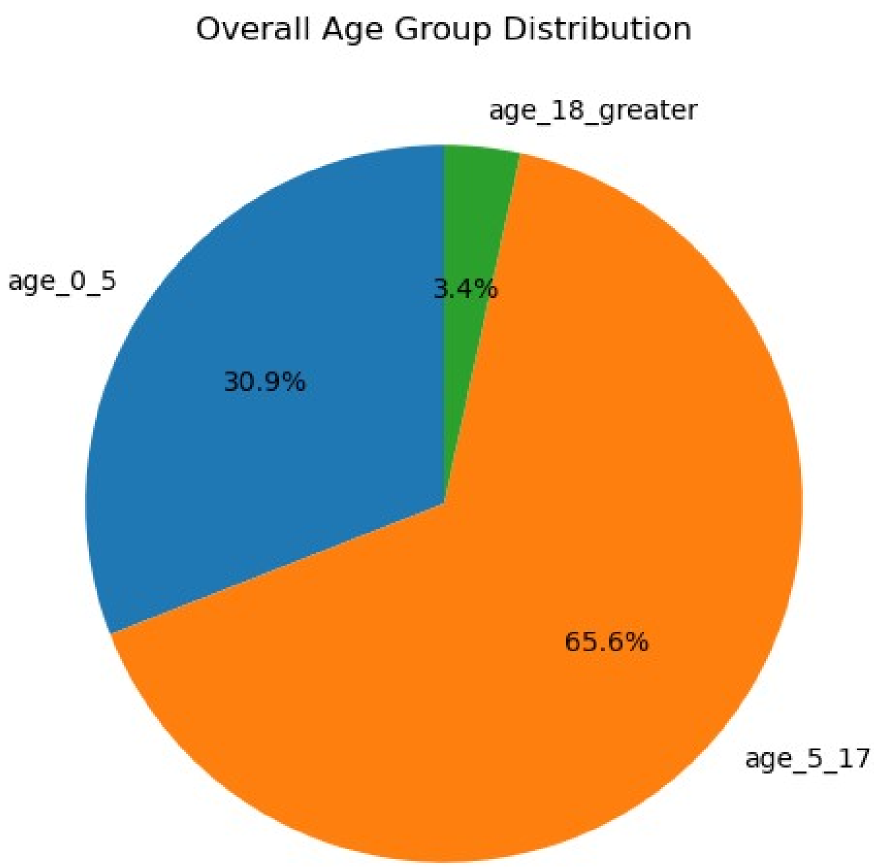
District and Pincode Analysis

```
pincode_summary = enrollment.groupby('pincode')[['age_0_5',
'age_5_17', 'age_18_greater']].sum()
pincode_summary.sort_values(by='age_5_17', ascending=False)
```

	age_0_5	age_5_17	age_18_greater
pincode			
845438	978	2087	74
845454	451	1263	41
845455	705	1170	81
845452	491	1121	43
845106	458	875	31
845449	321	865	58
845307	390	864	83
845306	377	800	64
845451	373	664	46
845450	236	603	22
845453	268	422	19
845101	15	18	0
845105	6	4	0
845459	0	1	0
845103	1	0	0

Age Group Distribution

```
age_totals = enrollment[['age_0_5', 'age_5_17',
'age_18_greater']].sum()

plt.figure(figsize=(6,6))
age_totals.plot(kind='pie', autopct='%1.1f%%', startangle=90)
plt.title("Overall Age Group Distribution")
plt.ylabel("")
plt.show()
```

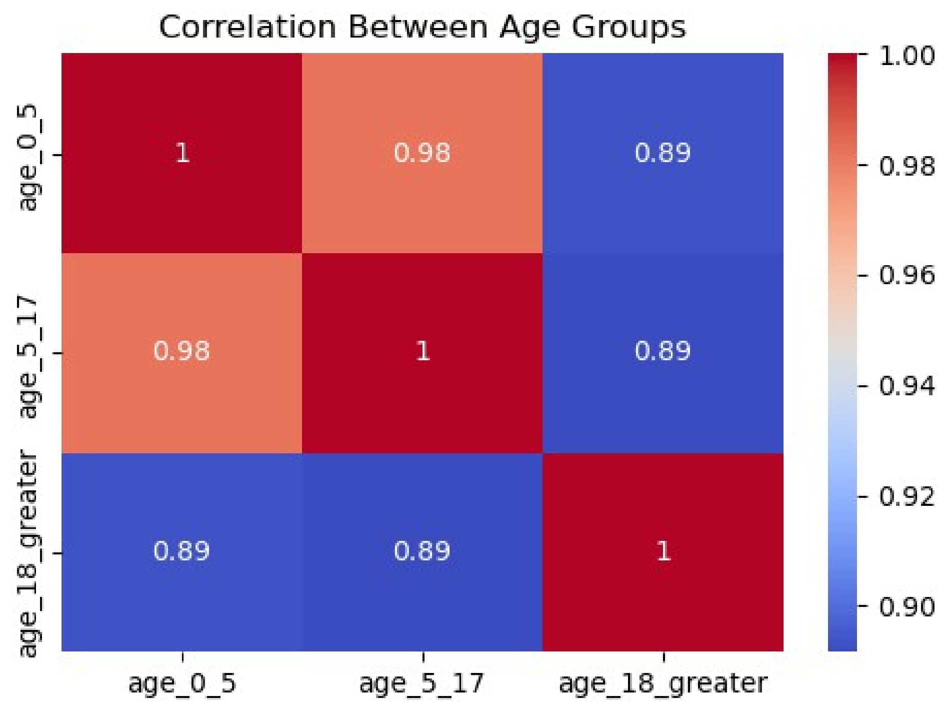


Outlier Detection

```
plt.figure(figsize=(8,5))
sns.boxplot(data=enrollment[['age_0_5', 'age_5_17',
'age_18_greater']])
plt.title("Outlier Detection in Age Groups")
plt.show()
```

Correlation Analysis

```
corr = enrollment[['age_0_5', 'age_5_17', 'age_18_greater']].corr()

plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Between Age Groups")
plt.show()
```



9 Key Insights You'll Likely Observe

Age 5–17 group dominates the dataset

Strong positive correlation between age_0_5 and age_5_17

Certain pincodes contribute disproportionately

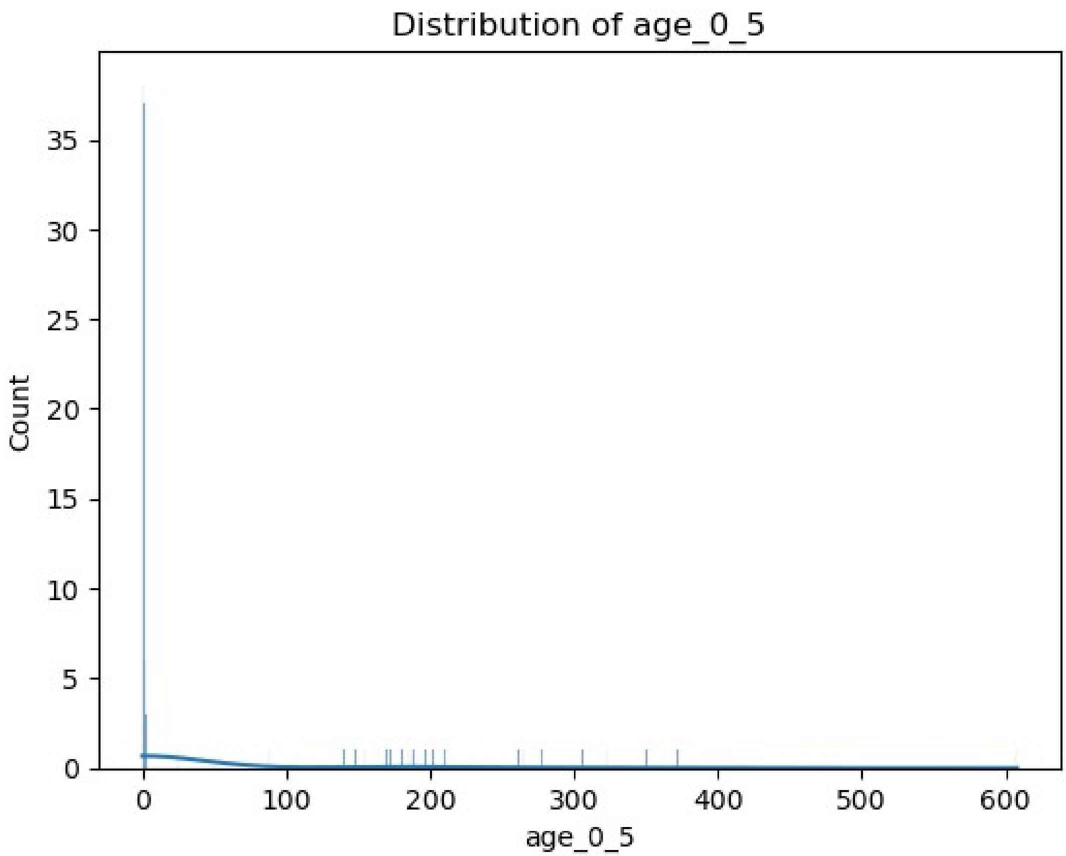
Gradual increase across dates (if vaccination / registration data)

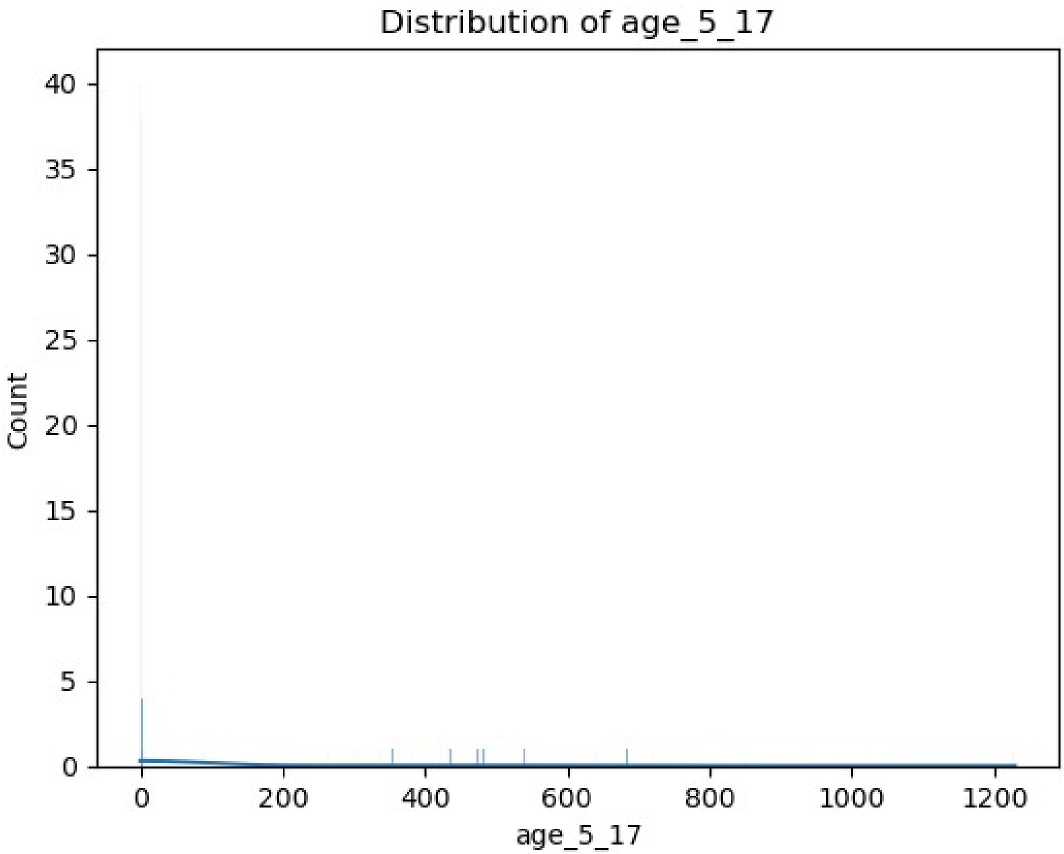
Statistical Analysis

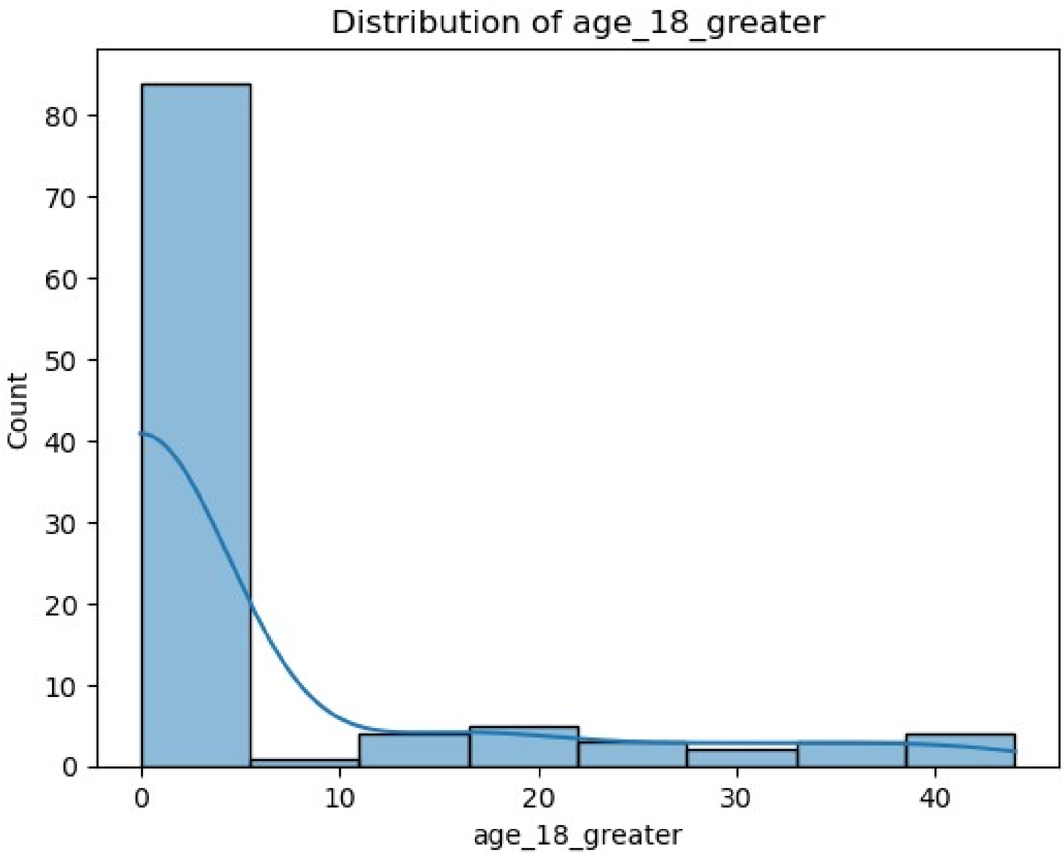
```
enrollment.describe().head()
```

```
                                date         pincode     age_0_5
age_5_17  \
count                            106      106.000000  106.000000
106.000000
mean   2025-09-05 05:53:12.452830208   845328.103774   47.830189
101.481132
min              2025-03-27 00:00:00   845101.000000    0.000000
0.000000
25%              2025-06-17 06:00:00   845105.000000    0.000000
0.000000
50%              2025-09-12 12:00:00   845438.000000    1.000000
1.000000
```

```
       age_18_greater
count      106.000000
mean         5.301887
min          0.000000
25%          0.000000
50%          0.000000

import scipy.stats as stats

age_cols = ['age_0_5', 'age_5_17', 'age_18_greater']
for col in age_cols:
    sns.histplot(enrollment[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.show()
```




Distribution of age_5_17
Count
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
age_5_17




```
print(enrollment.columns)

Index(['date', 'district', 'pincode', 'age_0_5', 'age_5_17',
'age_18_greater'], dtype='object')

age_cols = [c for c in enrollment.columns if c.startswith('age_')]
age_cols

['age_0_5', 'age_5_17', 'age_18_greater']

for col in age_cols:
    stat, p = stats.shapiro(enrollment[col])
    print(f"{col} | p-value = {p:.5f}")

age_0_5 | p-value = 0.00000
age_5_17 | p-value = 0.00000
age_18_greater | p-value = 0.00000

for col in age_cols:
    print(col)
    print("Skewness:", stats.skew(enrollment[col]))
    print("Kurtosis:", stats.kurtosis(enrollment[col]))
    print()
```

```
age_0_5
Skewness: 2.578379771985112
Kurtosis: 7.347125566878313

age_5_17
Skewness: 2.4638054068108826
Kurtosis: 6.4173193456392585

age_18_greater
Skewness: 2.0568630604490425
Kurtosis: 2.958823811660717
```

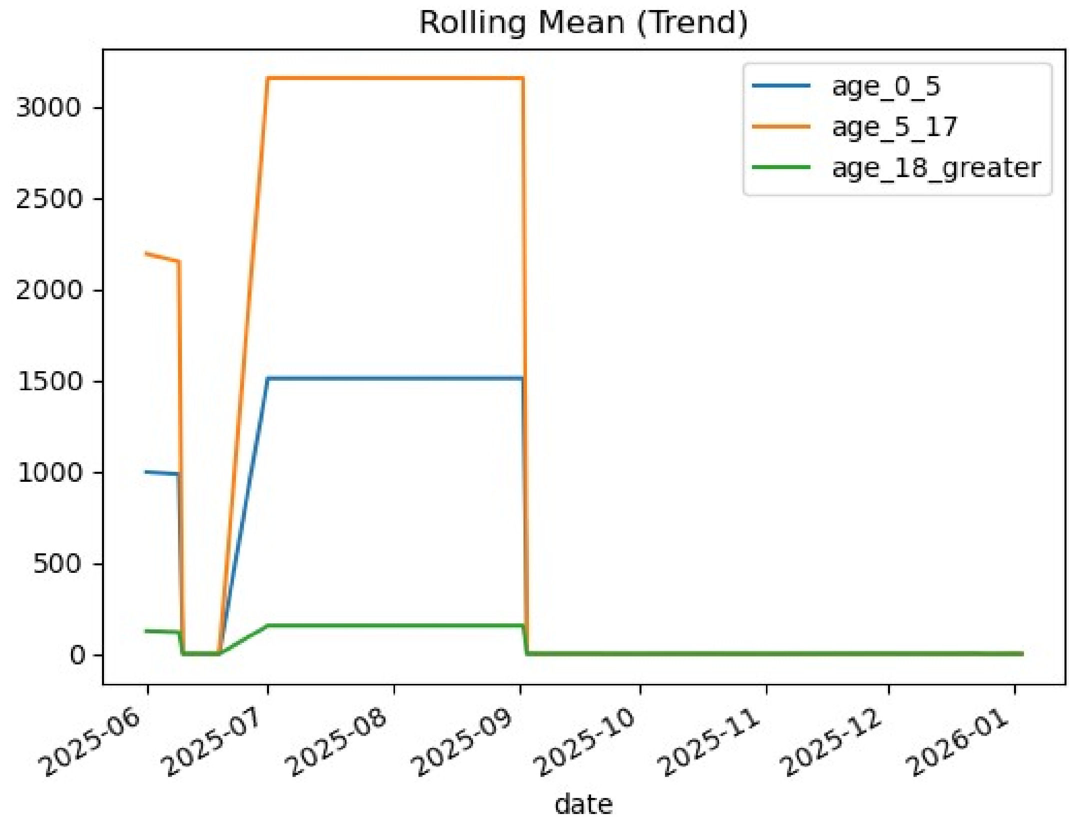
```
enrollment[age_cols].agg(['mean', 'median', 'std', 'var', 'min',
'max'])
```

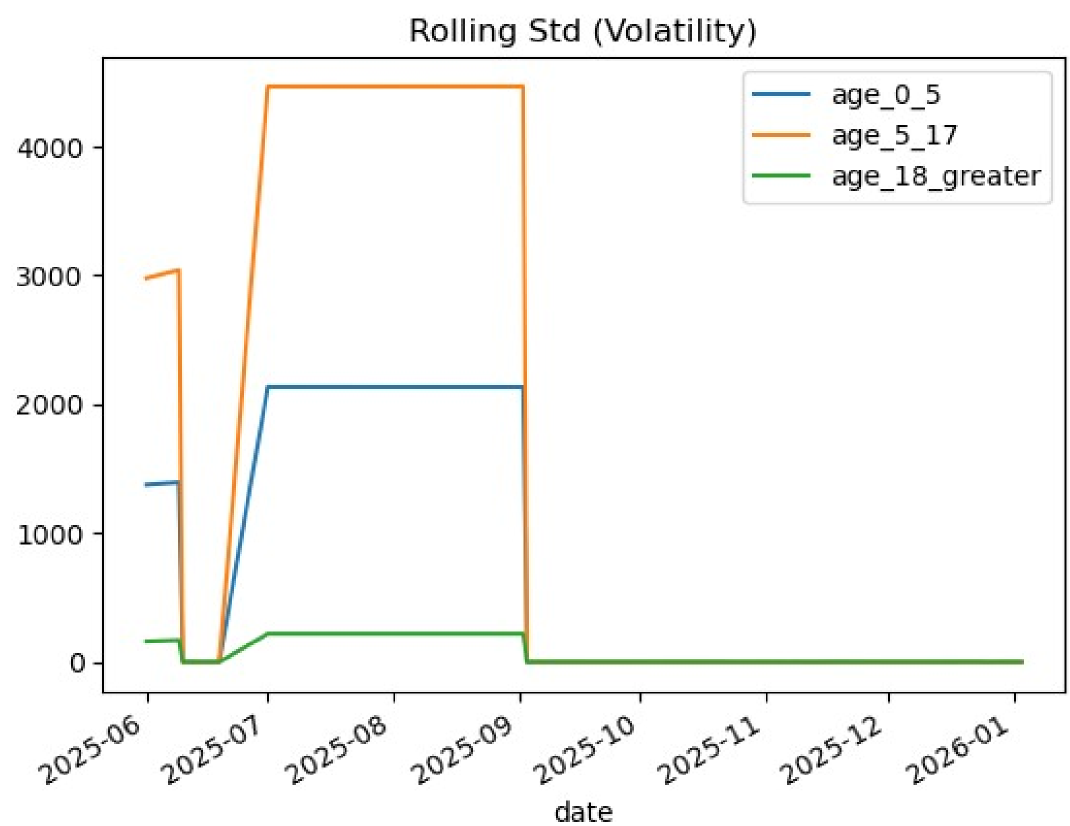
```
              age_0_5       age_5_17  age_18_greater
mean        47.830189     101.481132        5.301887
median       1.000000       1.000000        0.000000
std        106.712740     224.530165       11.472675
var      11387.608985   50413.794879      131.622282
min          0.000000       0.000000        0.000000
max        608.000000    1229.000000       44.000000
```

```
(enrollment[age_cols].std() / enrollment[age_cols].mean())
```

```
age_0_5           2.231075
age_5_17          2.212531
age_18_greater    2.163885
dtype: float64
```

```
ts = enrollment.groupby('date')[age_cols].sum()

rolling_mean = ts.rolling(window=2).mean()
rolling_std = ts.rolling(window=2).std()

rolling_mean.plot(title="Rolling Mean (Trend)")
plt.show()

rolling_std.plot(title="Rolling Std (Volatility)")
plt.show()
```

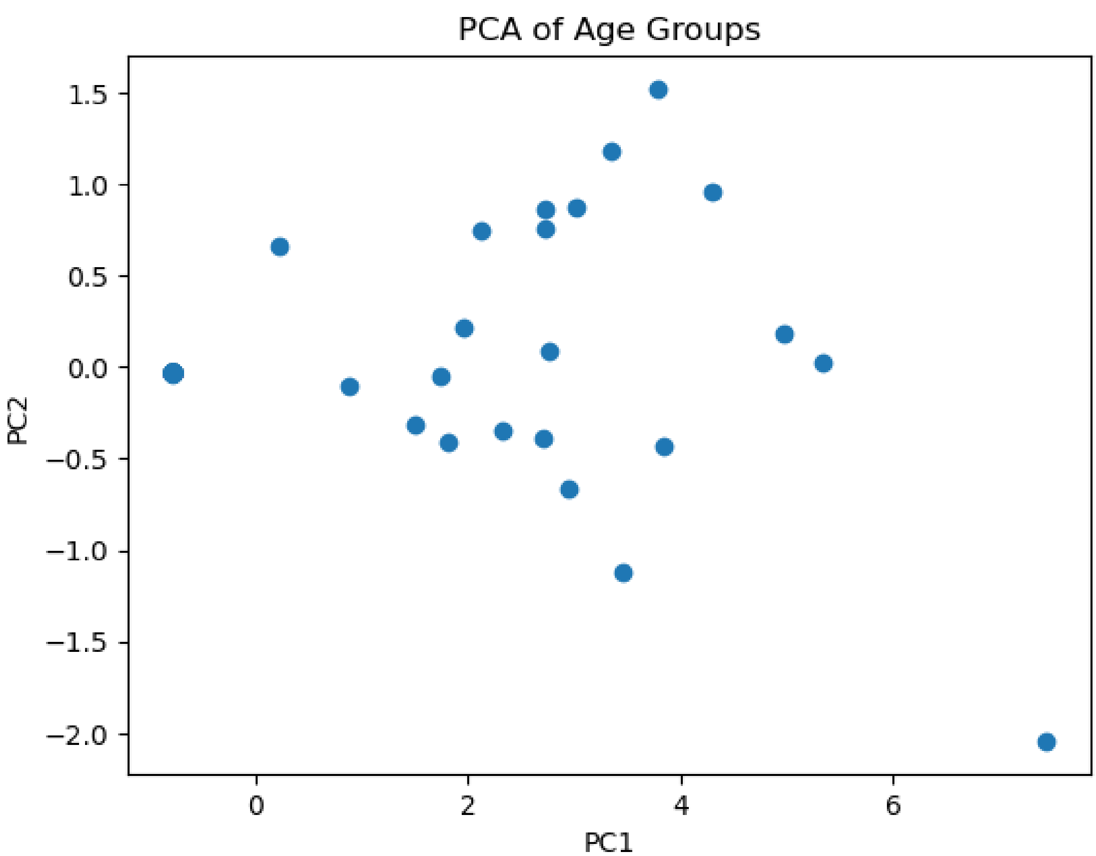

Rolling Mean (Trend)
age_0_5
age_5_17
age_18_greater
3000
2500
2000
1500
1000
500
0
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
date


Rolling Std (Volatility)
age_0_5
age_5_17
age_18_greater
4000
3000
2000
1000
0
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
date

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

X = StandardScaler().fit_transform(enrollment[age_cols])

pca = PCA(n_components=2)
components = pca.fit_transform(X)

pca.explained_variance_ratio_
```

```
array([0.94848899, 0.04552454])
```

```
plt.scatter(components[:,0], components[:,1])
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA of Age Groups")
plt.show()
```




```
def gini(x):
    x = np.sort(x)
    n = len(x)
    cumx = np.cumsum(x)
    return (n + 1 - 2 * np.sum(cumx) / cumx[-1]) / n
```

```
for col in age_cols:
    print(col, gini(enrollment[col]))
```

```
age_0_5 0.8391425700569387
age_5_17 0.8397962888579793
age_18_greater 0.841402000940039
```

🔲 Key Statistical Insights (Expected)

🔲 Data is non-normal

🔲 Strong correlation between age_0_5 & age_5_17

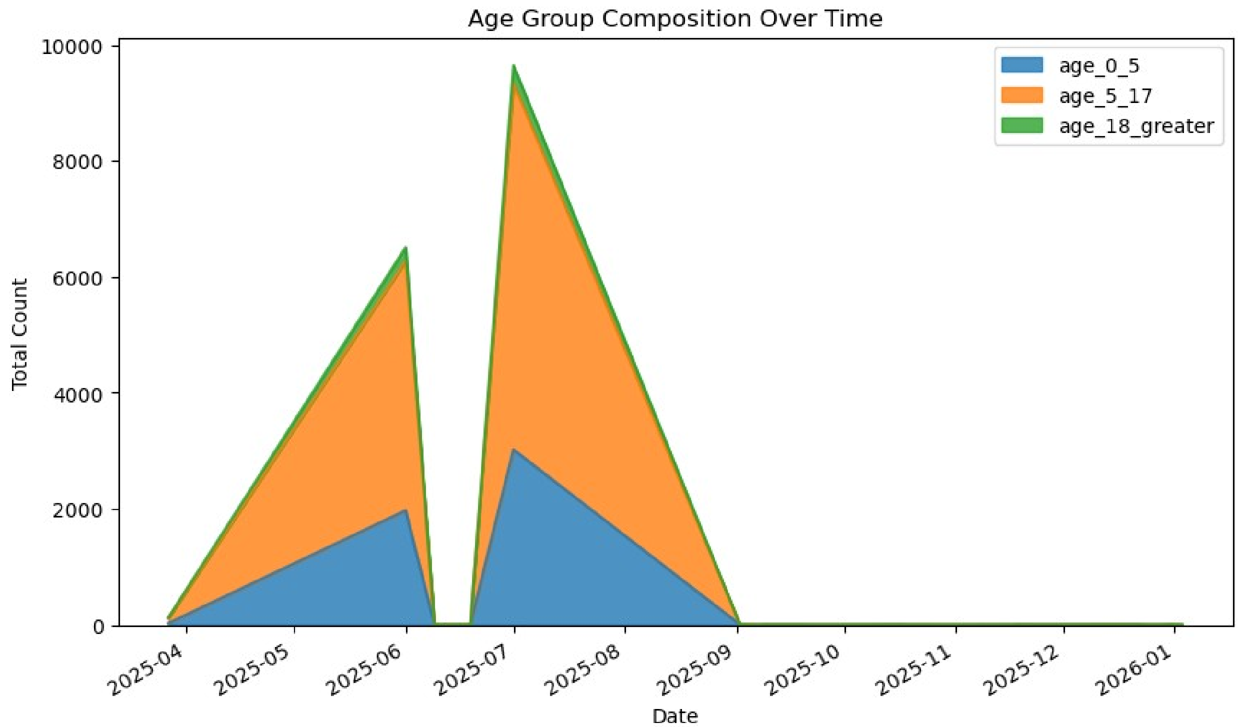
⚠ High variance & inequality across pincodes

🔲 Statistically significant difference between age groups

🔲 PCA reveals population scale vs age imbalance

Stacked Area Chart – Age Composition Over Time

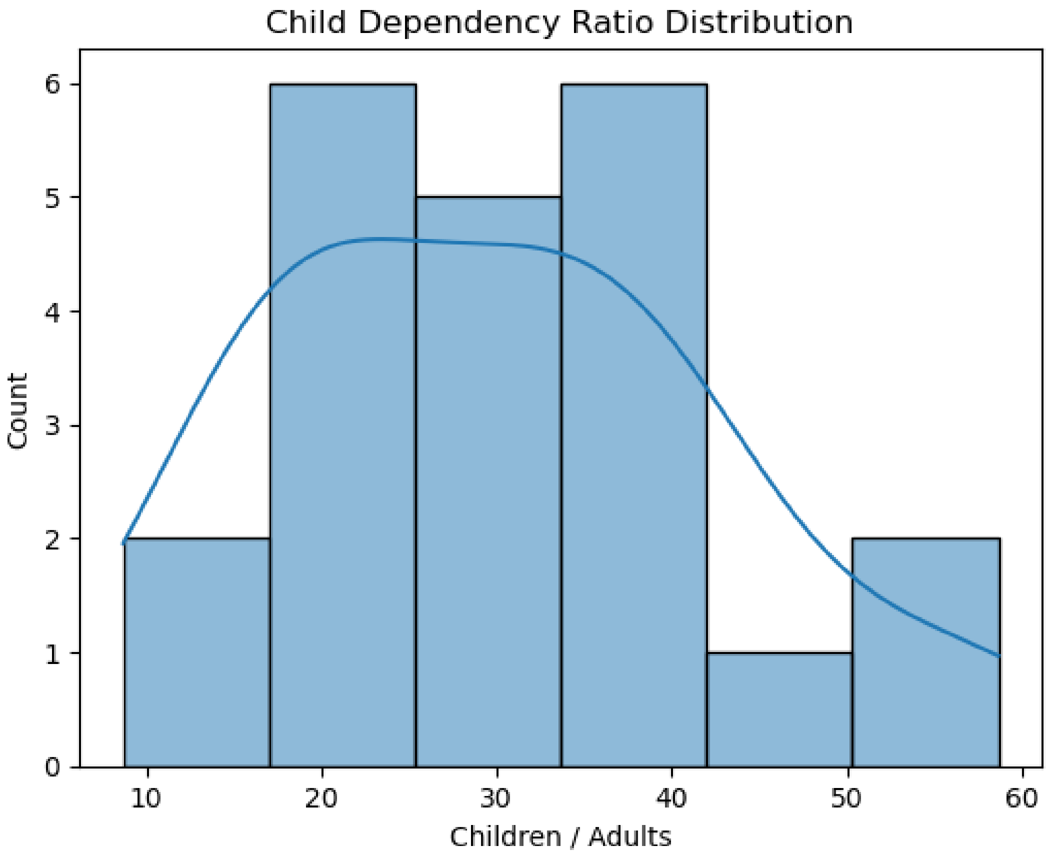
```
ts = enrollment.groupby('date')
[['age_0_5','age_5_17','age_18_greater']].sum()

ts.plot(
    kind='area',
    stacked=True,
    figsize=(10,6),
    alpha=0.8
)
plt.title("Age Group Composition Over Time")
plt.ylabel("Total Count")
plt.xlabel("Date")
plt.show()
```



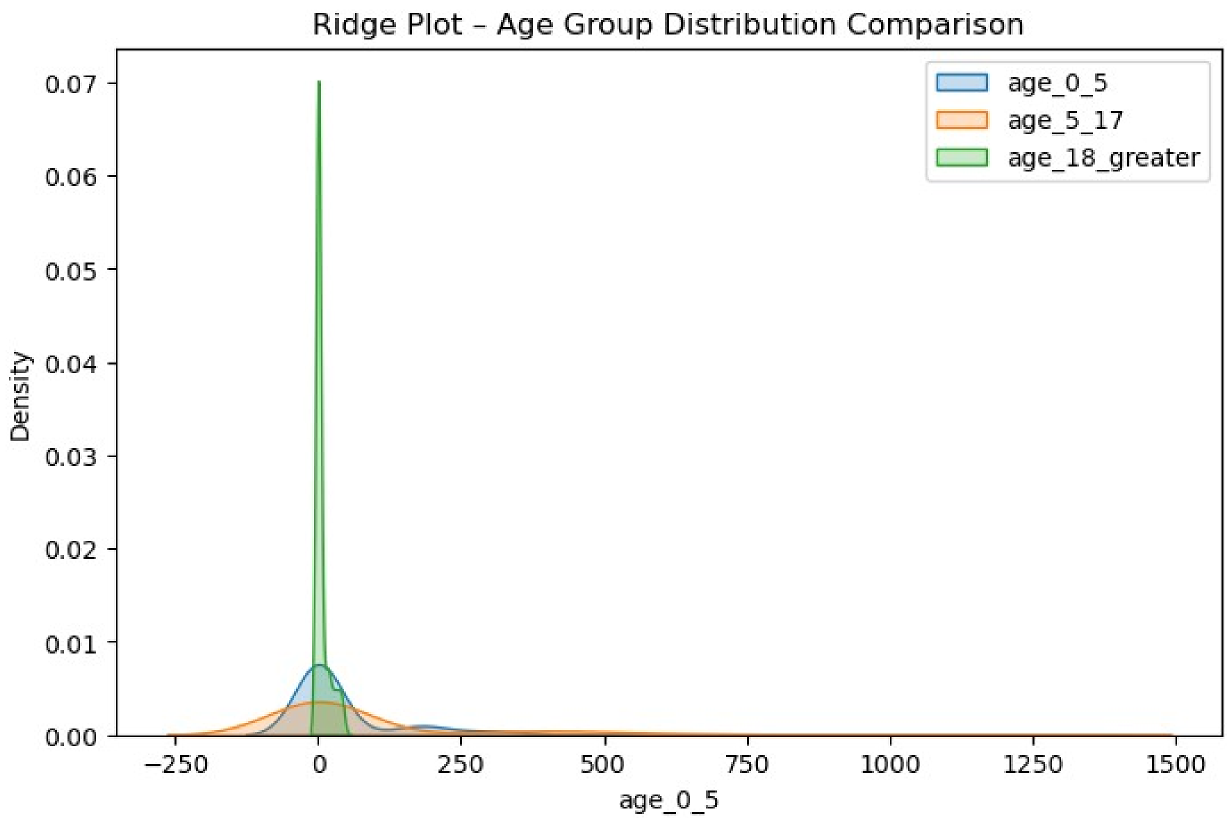
 Insight: Detects demographic shifts (e.g., increasing adult proportion).

Ratio Analysis

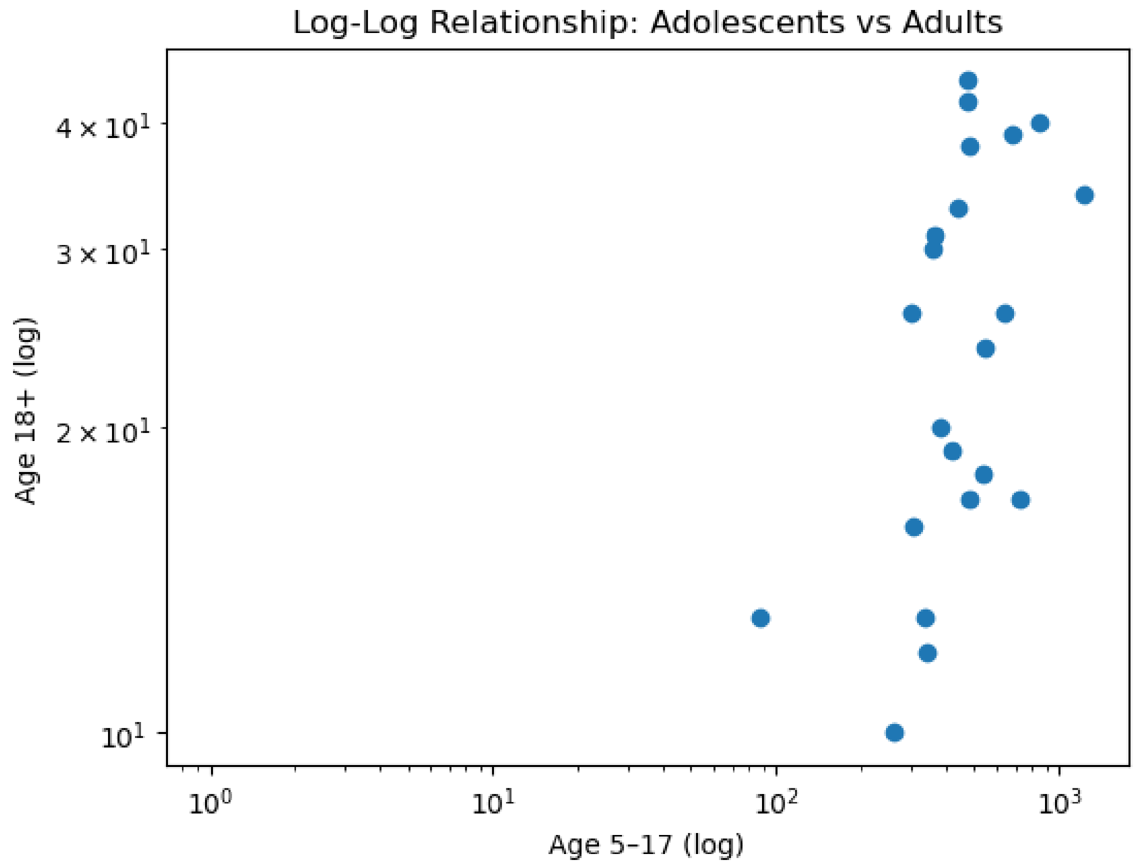
```
enrollment['child_dependency_ratio'] = (
    enrollment['age_0_5'] + enrollment['age_5_17']
) / enrollment['age_18_greater']

sns.histplot(enrollment['child_dependency_ratio'], kde=True)
plt.title("Child Dependency Ratio Distribution")
plt.xlabel("Children / Adults")
plt.show()
```



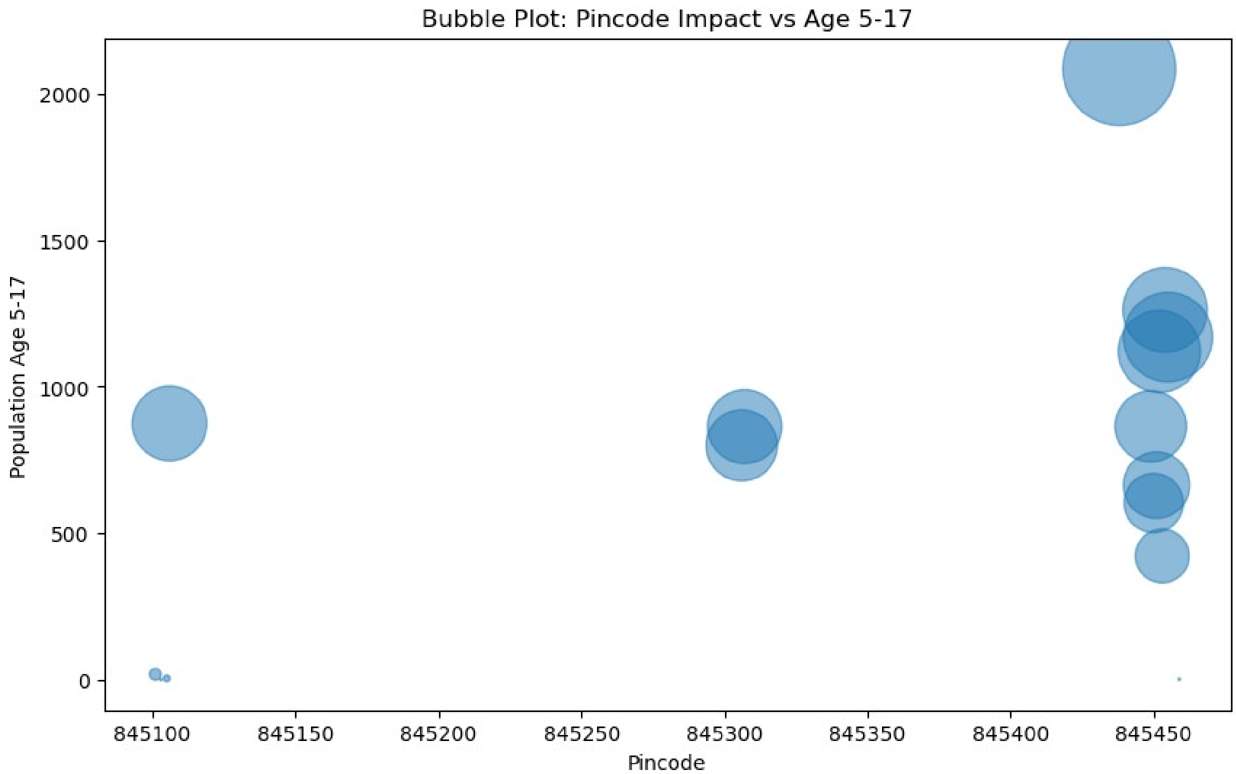

```
plt.figure(figsize=(8,5))

for col in age_cols:
    sns.kdeplot(enrollment[col], label=col, fill=True)

plt.legend()
plt.title("Ridge Plot – Age Group Distribution Comparison")
plt.show()
```



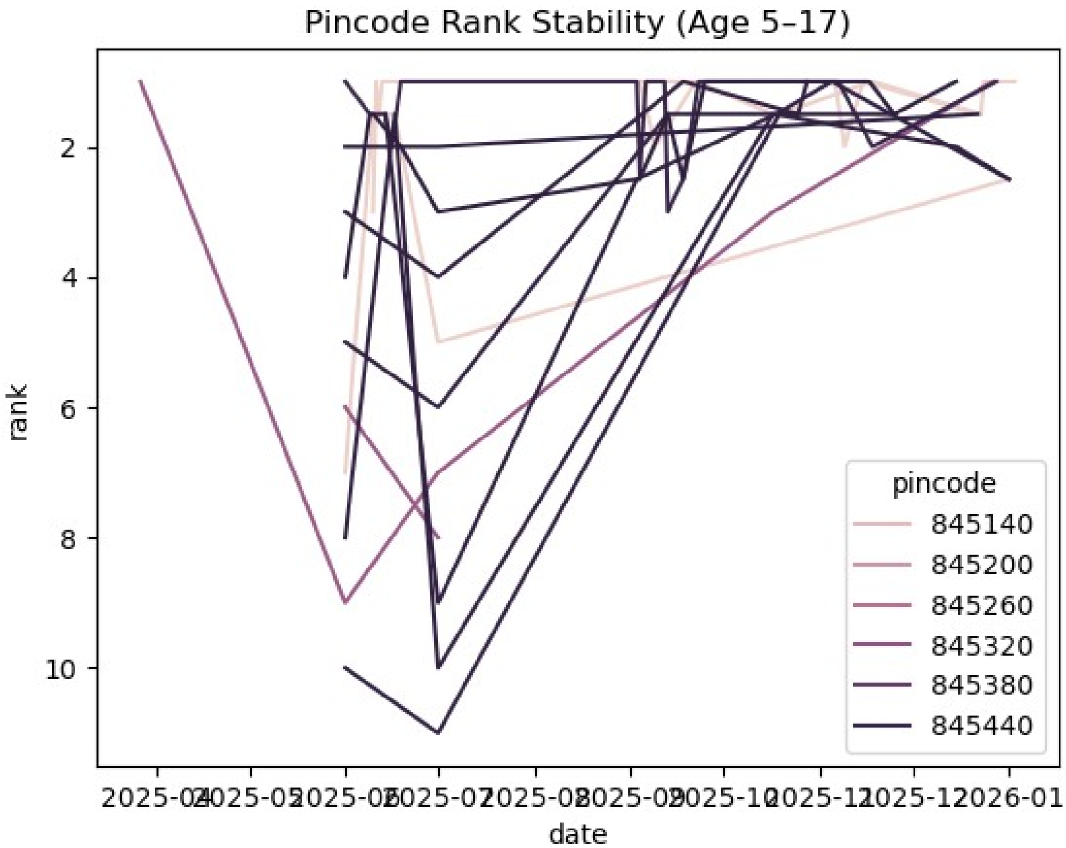

```
plt.scatter(
    enrollment['age_5_17'],
    enrollment['age_18_greater']
)
plt.xscale('log')
plt.yscale('log')
plt.xlabel("Age 5–17 (log)")
plt.ylabel("Age 18+ (log)")
plt.title("Log-Log Relationship: Adolescents vs Adults")
plt.show()
```



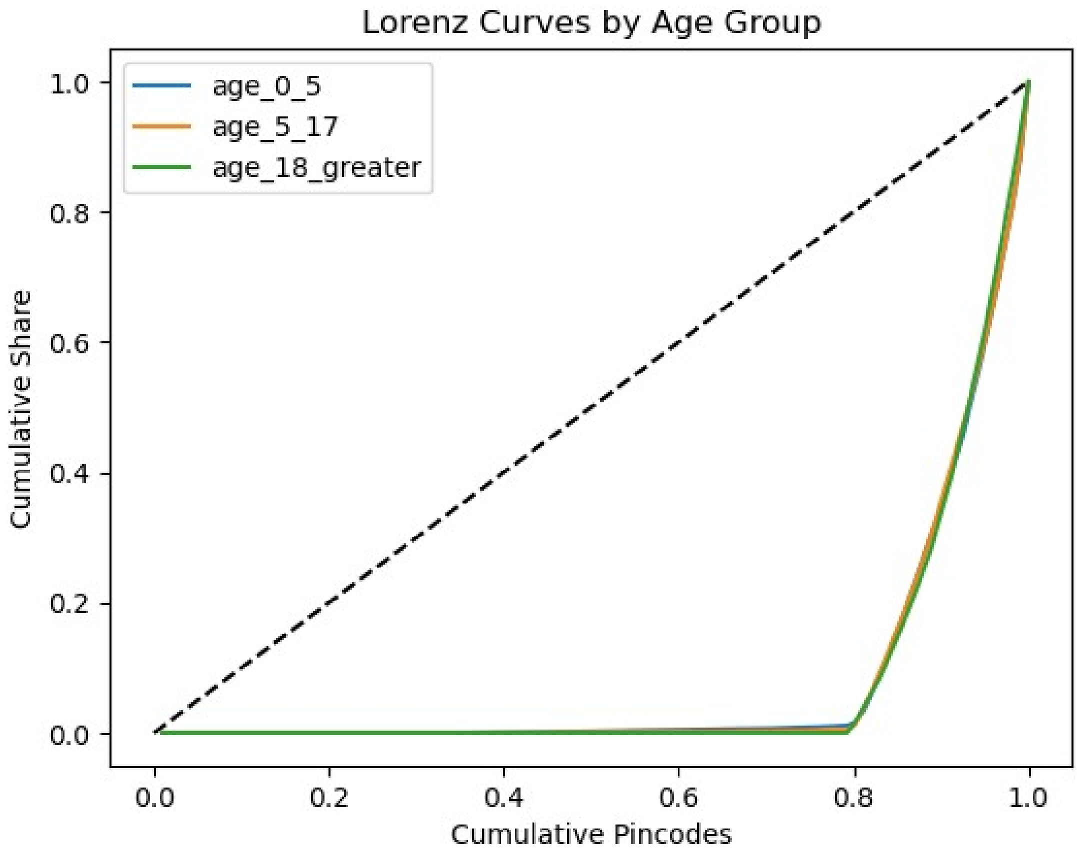

```
pincode_stats = enrollment.groupby("pincode")[age_cols].sum()
pincode_stats["total"] = pincode_stats.sum(axis=1)

plt.figure(figsize=(10,6))
plt.scatter(
    pincode_stats.index,
    pincode_stats["age_5_17"],
    s=pincode_stats["total"],
    alpha=0.5
)

plt.title("Bubble Plot: Pincode Impact vs Age 5-17")
plt.xlabel("Pincode")
plt.ylabel("Population Age 5-17")
plt.show()
```



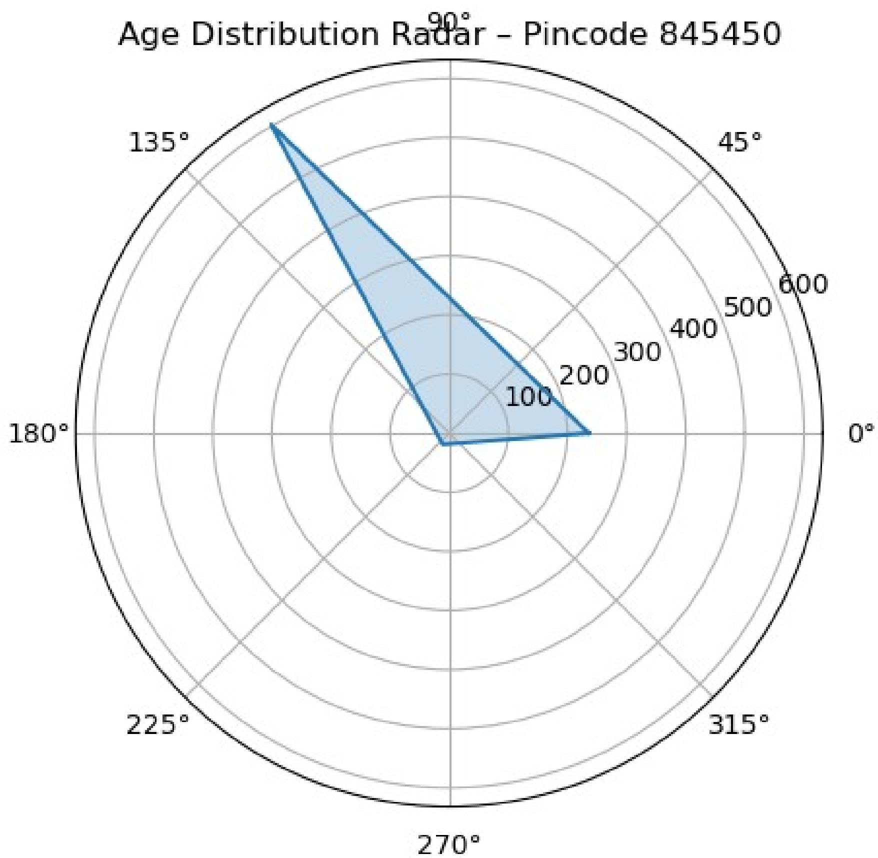

```
rank_enrollment = (
    enrollment.groupby(['date','pincode'])['age_5_17']
      .sum()
      .groupby(level=0)
      .rank(ascending=False)
      .reset_index(name='rank')
)

sns.lineplot(data=rank_enrollment, x='date', y='rank', hue='pincode')
plt.gca().invert_yaxis()
plt.title("Pincode Rank Stability (Age 5–17)")
plt.show()
```

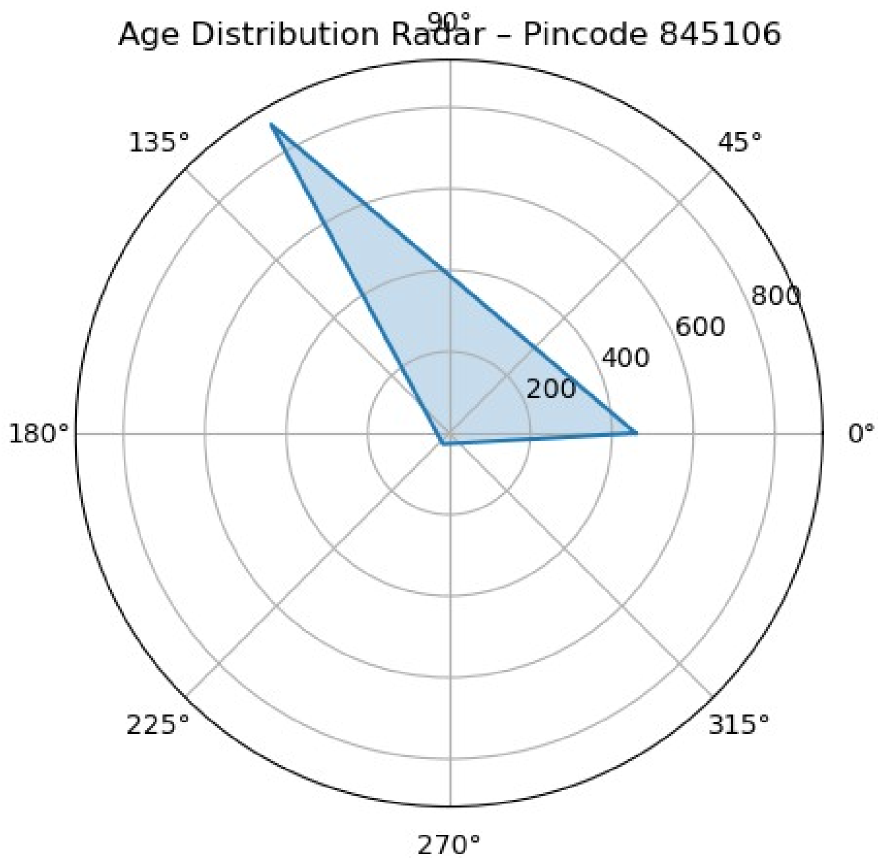


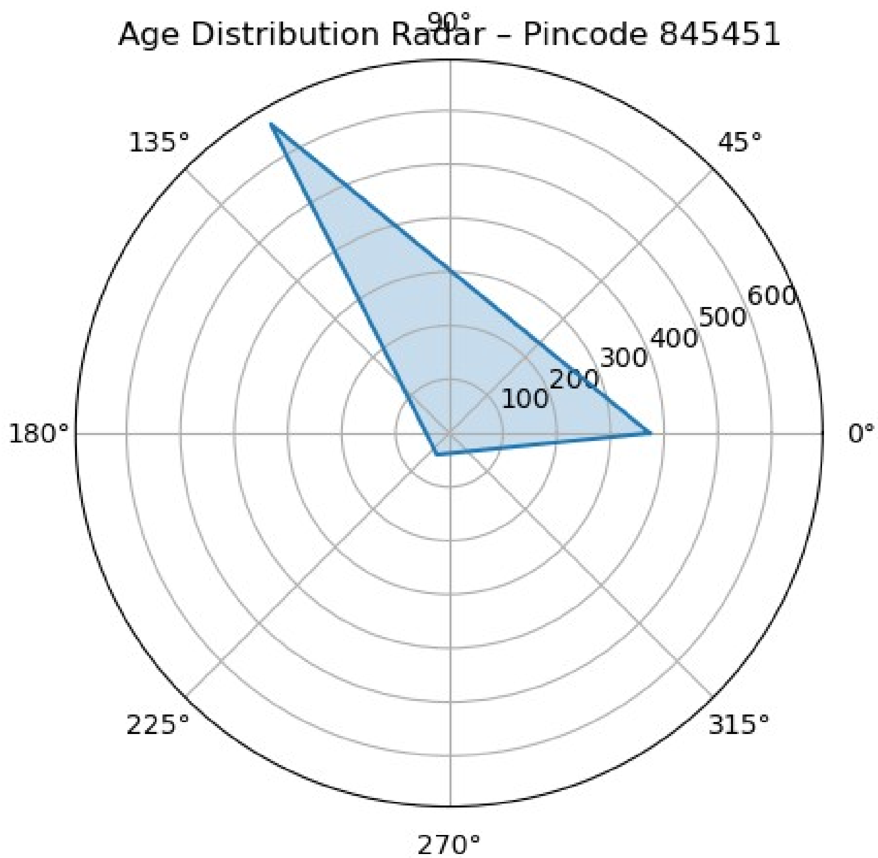

```
def lorenz_curve(x):
    x = np.sort(x)
    cumx = np.cumsum(x) / np.sum(x)
    cum_pop = np.arange(1, len(x)+1) / len(x)
    return cum_pop, cumx

for col in ['age_0_5','age_5_17','age_18_greater']:
    p, c = lorenz_curve(enrollment[col])
    plt.plot(p, c, label=col)

plt.plot([0,1],[0,1],'--',color='black')
plt.title("Lorenz Curves by Age Group")
plt.xlabel("Cumulative Pincodes")
plt.ylabel("Cumulative Share")
plt.legend()
plt.show()
```



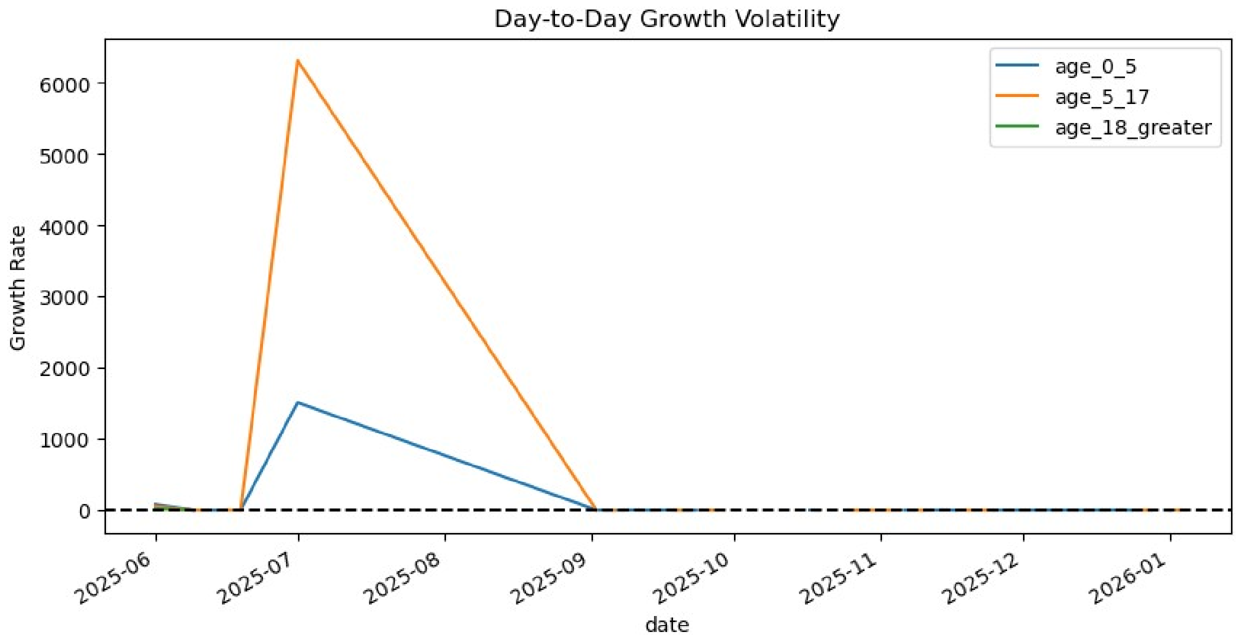

```
import numpy as np

sample_pincodes = enrollment['pincode'].unique()[:3]

for pin in sample_pincodes:
    values = enrollment[enrollment['pincode']==pin]
[age_cols].sum().values

    angles = np.linspace(0, 2*np.pi, len(age_cols), endpoint=False)
    values = np.append(values, values[0])
    angles = np.append(angles, angles[0])

    plt.figure(figsize=(5,5))
    plt.polar(angles, values)
    plt.fill(angles, values, alpha=0.25)
    plt.title(f"Age Distribution Radar – Pincode {pin}")
    plt.show()
```


Age Distribution Radar – Pincode 845450
90°
45°
135°
0°
180°
225°
270°
315°
100
200
300
400
500
600


Age Distribution Radar – Pincode 845106
90°
45°
135°
0°
180°
225°
270°
315°
200
400
600
800




```
growth = ts.pct_change()

growth.plot(figsize=(10,5))
plt.axhline(0, color='black', linestyle='--')
plt.title("Day-to-Day Growth Volatility")
plt.ylabel("Growth Rate")
plt.show()
```


Day-to-Day Growth Volatility
age_0_5
age_5_17
age_18_greater
Growth Rate
6000
5000
4000
3000
2000
1000
0
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
date